For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

FURTHER MATHEMATICS

PAPER 2

9187/2

NOVEMBER 2013 SESSION

3 hours

Additional materials:
- Answer paper
- Graph paper
- List of Formulae

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

There is no restriction on the number of questions which you may attempt.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

If a numerical value for g is necessary, take $g = 9.81 \text{ ms}^{-2}$.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 120.

Within each section of the paper, questions are printed in the order of their mark allocations and candidates are advised, within each section, to attempt questions sequentially.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 7 printed pages and 1 blank page.

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Section (a) : Mechanics

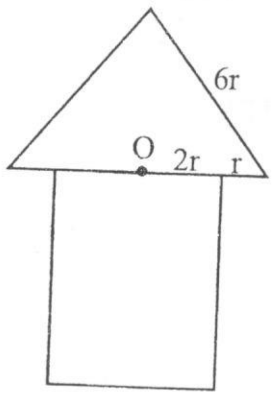
1 A conical pendulum consists of a light inextensible string AB, fixed at A and carrying a particle of mass 0.05 kg at B.
The particle moves in a horizontal circle of radius $\sqrt{2}$ m and centre of gravity below A. If the angle between the string and the vertical is 30°,

find

(i) the tension in the string, [2]

(ii) the angular speed of the particle. [2]

2 An African hut model is in the form of a solid uniform cylinder joined to a solid cone. The diagram shows the cross section of the African hut model.

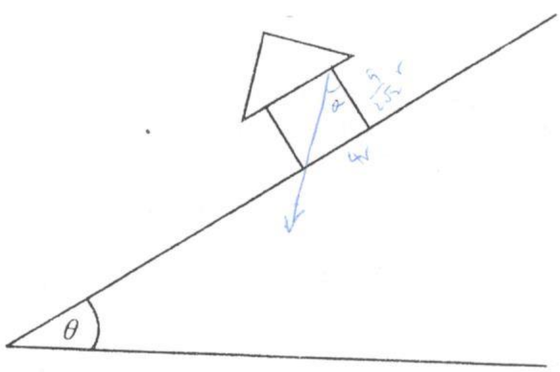


Sin 30
cos 30 = x/1

The radius of the cylinder is $2r$ and that of the cone is $3r$ and is of slant length $6r$. The centre of mass of the model is at O (see diagram).

(a) Show that the height of the cylinder is $\frac{9}{2\sqrt{2}}r$. [6]

(b) The hut model is now placed on a rough inclined plane as shown below.



$F = mr\omega^2$

Given that the model is at the point of toppling, find the angle, θ, which the inclined plane makes with the horizontal. [2]

3 A light elastic spring of natural length l m and modulus of elasticity $10g$ N, hangs vertically with its upper end fixed and a body of mass M kg attached to its lower end. The body initially rests in equilibrium and is then pulled down a distance of $\frac{l}{2}$ m and then released.

(a) Show that the resulting motion will be simple harmonic. [4]

(b) Given that $l = 1$ m and $M = 8$ kg, find the period of motion and its amplitude. [3]

(c) Find the maximum speed of the body. [2]

$P = Fv$

$P = Fv$

4 A recovery vehicle of mass 1.2 tonnes tows a bus of mass 3.6 tonnes along a straight horizontal road. The recovery vehicle's engine is working at a steady rate of 50 kW and the total resistance to motion is 2 000 N.

(a) Show that when the recovery vehicle is moving at 60 km/h, its acceleration is 0.21 correct to 2 significant figures. [4]

$\frac{60\,km}{hr} \times \frac{1000}{3600}$

(b) Find the maximum speed of the recovery vehicle. [2] $F = 833.33$

(c) Assuming that the total resistance force is divided equally between the recovery vehicle and the bus, find the tension in the coupling when the speed is 60 km/h. [3]

2000

$F - R = ma$

$833.33 - 2000 =$

5 Two spheres A and B are on a rough horizontal table where the coefficient of friction between each sphere and the surface is 0.2. Spheres A and B have masses 2 kg and 1 kg respectively. Sphere A is 2 m from sphere B. The spheres A and B are then given initial speeds of 10 m/s and 1 m/s respectively in the same direction with sphere A following sphere B.

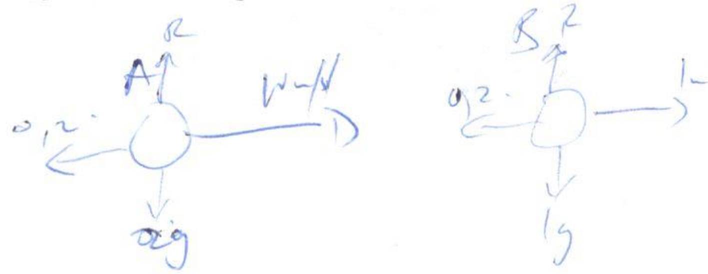
(i) Show that both spheres decelerate at the same rate. [2]

(ii) Find the time which elapses before collision takes place. [3]

(iii) Find the speed of sphere A and the speed of sphere B just before impact. [2]

$T = \frac{\lambda x}{l}$

(iv) On impact, the two spheres coalesce, find the speed of the combined spheres after impact. [2]



$F = \mu R$

$F =$

[Turn over

6 A man projects a block of mass 1 kg from a point A, up the line of greatest slope from the bottom of a rough plane inclined at 60° to the horizontal. The coefficient of friction between the block and the plane is 0.3. The block first comes to rest at point B which is 4 m vertically above the point of projection before it returns to A.

Find

(a) work done against friction when the body moves from A to B, [2]

(b) the initial speed of the body, [4]

(c) the work done against friction when the body moves from A to B and back to A, [1]

(d) the speed of the body on return to A. [3]

7 A uniform ladder AB of mass 8 kg and of length 6 m is resting in equilibrium at an angle of 50° to the horizontal, with its upper end A against a smooth vertical wall and its lower end B on a rough horizontal ground with coefficient of friction μ.

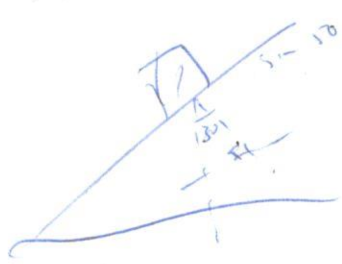
Find

(i) **1.** the normal reaction with the smooth wall and the normal reaction with the rough horizontal ground.

2. the frictional force F and hence show that the least possible value of μ is 0.42 correct to 2 decimal places. [6]

(ii) A cat of mass 3 kg climbs up the ladder AB.

Find how far up the ladder it can ascend before the ladder slips. [5]



$v^2 = u^2 + 2as$

$F - \mu R = ma$

Section (b): Statistics

8 If X is binomially distributed with $P(X=x) = {}^{n}C_{x}p^{x}q^{n-x}$ for $x = 0,1,2, \ldots n$ where $q = 1 - p$ and $0<p<1$.

(i) Show that the probability generating function G(t) is

$$G(t) = (q + pt)^{n},$$ [3]

(ii) Hence use the above results to show that the mean and variance of X are 2.2 and 0.99 respectively given that $n = 4$ and $p = 0.55$. [5]

9 Potatoes are packed into packets which the farmer claims are 10 kg each on average. A random sample of 100 bags were examined and the mass, x kg, of the contents of each packet was determined. It was found that

$\sum(x-10) = 101$ and $\sum(x-10)^{2} = 838$.

(a) Calculate the unbiased estimate of the population mean μ and show that the unbiased estimate of the population variance σ^{2} is 7.43 correct to 2 decimal places. [5]

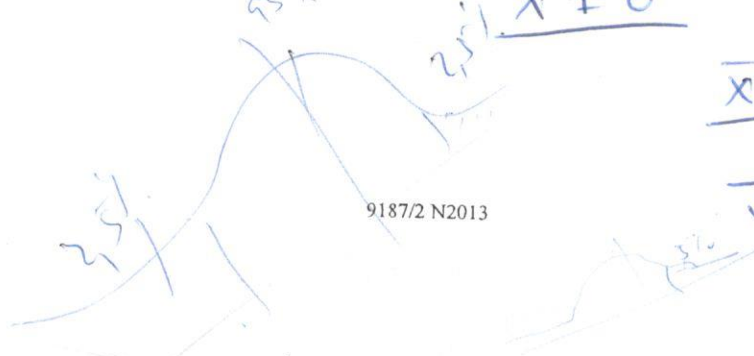
(b) Find a 90% confidence interval for which the mean mass lies. [3]

10 A manufacturer packs sugar into packets which he claims each weighs 1 kg on average. A random sample of 8 packets gave the following results 1.01 1.02 0.99 1.00 1.01 0.98 1.03 1.01

Assuming that the masses of the contents of packets are normally distributed,

(i) find the mean mass of each packet and the standard deviation, [3]

(ii) test at 5% level of significance, the hypothesis that the average mass is 1 kg against the alternative hypothesis that it is not 1 kg. [6]



$\bar{x} \pm \sigma$ $\bar{x} + \mu$

$\dfrac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$

11 The number of errors made per page by a copy typist has a Poison distribution with mean 0.2.

Calculate correct to 2 decimal places, the probability that

(i) the first page typed will contain no error, [2]

(ii) the total number of errors in the first two pages is 2 or less, [3]

(iii) the first error will appear on the third page she types, [1]

(iv) exactly 2 of the first 5 pages will contain no error. [3]

12 The number of accidents occurring each week at Hwanda Bridge were recorded over a period of 100 weeks. The results are tabulated below.

number of accidents	frequency
0	5
1	23
2	23
3	25
4	14
5	10
6 or more	0

(a) Find the mean number of accidents per week. [1]

(b) Test at 5% level of significance whether the number of accidents follow a poison distribution. [12]

$E = f\,(\,P(x=r)$

13 A random sample of 10 families yielded the following results on monthly income (x) and savings (y).

Family	**A**	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**
Income (x)	150	230	280	180	200	130	220	250	100	150
Savings (y)	10	25	30	20	15	12	20	28	12	15

(a) Plot a scatter diagram and comment on the nature of relationship between income and savings.

[5]

(b) **(i)** Find the regression equation of savings (y) on income (x).

(ii) Hence predict the savings for a family whose income is \$240.

(iii) Using the equation in **b(i)**, find out with reasons, whether it is possible to predict the level of savings for a family whose monthly income is \$300.

[8]